

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10 - 1.1 PRESSURE: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

You have finished all 9 lessons on pressure. Now you will write your own Final Explanation of the driving question: Why does a sharp knife cut better than a blunt knife when using the same force? Answer each section in the space below it. Use numbers with their units, and use evidence from your lessons and your Driving Question Board. Use $g = 10 \text{ N/kg}$. Remember to say what is the same and what is different between the two knives. Write in full sentences.

Part 1: Claim and Evidence from the Kitchen

Mama Wanjiru in Kisumu pushes with the same force on a sharp knife (edge 0.1 mm thick) and a blunt knife (edge 1 mm thick). Both knives are the same size and shape and touch the food along a length of 5 cm. (a) Write a claim that answers the driving question. (b) List what is the same and what is different in this situation. (c) Give the force value and what happened with each knife.

(a) Claim: The sharp knife cuts and the blunt knife only squashes because the sharp edge touches the food over a much smaller area, so it gives a much greater pressure. Same force, different area. The difference between the knives is the area, not the force. (b) Same: the person (Mama Wanjiru), the force of 40 N, the food (tomato, nyama choma), and the size and shape of the knives. Different: the edge thickness (0.1 mm for the sharp knife, 1 mm for the blunt knife) and the result. (c) Both knives were pushed with 40 N. The sharp knife sliced easily through the tomato and nyama choma. The blunt knife barely cut and only squashed the food.)

Part 2: Calculating the Pressure Under Each Knife

Use $P = F/A$ to explain the knives with numbers. (a) Change each edge thickness to metres and find the contact area of each knife (contact length 0.05 m). (b) Calculate the pressure under each knife and give your answers in MPa. (c) Compare the two areas and the two pressures. How many times

(a) Sharp knife: $0.1 \text{ mm} = 0.0001 \text{ m}$, so $A = 0.05 \times 0.0001 = 0.000005 \text{ m}^2 = 5 \times 10^{-6} \text{ m}^2$. Blunt knife: $1 \text{ mm} = 0.001 \text{ m}$, so $A = 0.05 \times 0.001 = 0.00005 \text{ m}^2 = 5 \times 10^{-5} \text{ m}^2$. (b) Sharp: $P = F/A = 40 / 0.000005 = 8,000,000 \text{ Pa} = 8 \text{ MPa}$. Blunt: $P = 40 / 0.00005 = 800,000 \text{ Pa} = 0.8 \text{ MPa}$. (c) The sharp edge has 10 times less area than the blunt edge, and it gives 10 times more pressure (8 MPa compared with 0.8 MPa). For a constant force, a smaller area gives a larger pressure. (d) The sharp knife's pressure of 8 MPa is greater than 2 MPa, so the tomato and meat give way and are cut. The blunt knife's pressure of 0.8 MPa is less than 2 MPa, so the food does not give way and is only squashed.)

larger or smaller is each? (d) Tomato skin and nyama choma give way at about 2 MPa. Use this value to explain what happens with each knife.	
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Part 3: Testing the Idea with Other Evidence

Use evidence from your earlier lessons to show that the area-pressure idea works beyond the knives. (a) In the brick-on-flour test, the same 2 kg brick (weight 20 N) was placed on its large face (0.02 m ²) and its small face (0.005 m ²). Calculate the pressure for each face and say which face sank deepest. (b) A camel (4,000 N, total foot area 0.2 m ²) and a 60 kg student (600 N, total shoe area 0.04 m ²) stand on sand. Calculate each pressure and explain why the much heavier camel does not sink much more than the student.	(a) Large face: $P = 20 / 0.02 = 1,000 \text{ Pa}$. Small face: $P = 20 / 0.005 = 4,000 \text{ Pa}$. The weight (20 N) is the same on each face, but the small face has the smaller area and the greater pressure, so it made the deepest dent in the flour. (b) Camel: $P = 4,000 / 0.2 = 20,000 \text{ Pa} = 20 \text{ kPa}$. Student: $P = 600 / 0.04 = 15,000 \text{ Pa} = 15 \text{ kPa}$. The camel is about 6.7 times heavier (4,000 N compared with 600 N), but its weight is spread over a larger area, so its pressure (20 kPa) is only a little larger than the student's (15 kPa). This is the same idea as the knives: pressure depends on force and area together, not on force alone.)
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Part 4: Pressure in Fluids and Its Uses

The knife idea can be extended to fluids. (a) Water in a dam is 30 m deep. Use $P = \rho gh$ (water density 1,000 kg/m ³) to calculate the pressure at the base and at 10 m depth. Explain why a dam wall is built thicker at the bottom. (b) In the glass-with-cardboard test, the glass mouth has area 0.004 m ² , the atmospheric pressure is 101 kPa, and the water weighs 2.5 N. Calculate the upward force of the air and explain what it shows. (c) In a hydraulic jack, a	(a) At the base: $P = \rho gh = 1,000 \times 10 \times 30 = 300,000 \text{ Pa} = 300 \text{ kPa}$. At 10 m: $P = 1,000 \times 10 \times 10 = 100,000 \text{ Pa} = 100 \text{ kPa}$. Liquid pressure increases with depth because there is more water weight above, so the wall must withstand 300 kPa at the base compared with 100 kPa higher up. That is why it is thicker at the bottom. (b) $F = P \times A = 101,000 \times 0.004 = 404 \text{ N}$ upward. This is much larger than the 2.5 N weight of the water, so the cardboard stays in place. Air has weight and so exerts atmospheric pressure on every surface. (c) Small piston area = 5 cm ² = 0.0005 m ² , so $P = 50 / 0.0005 = 100,000 \text{ Pa} = 100 \text{ kPa}$. This pressure is transmitted equally through the enclosed fluid (Pascal's principle). Large piston area = 100 cm ² = 0.01 m ² , so $F = 100,000 \times 0.01 = 1,000 \text{ N}$, which is 20 times 50 N. With the knives the force was the same and the area differed, giving different pressures. With the jack the pressure is the same and the area differs, giving different forces. In both cases, area is the key.)
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50 N effort acts on a piston of 5 cm ² and the large piston has 100 cm ² . Find the pressure and the load force, and say how the idea differs from the knives.	
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Part 5: Final Explanation and Model Revision

Write your Final Explanation of the driving question as a claim, evidence and reasoning paragraph. Use the numbers from the knives. Then look back at your first model from Lesson 1: say what you had right, what you had to add or change, and which lessons helped you do this.	Claim: The sharp knife cuts better than the blunt knife because its thin edge has a smaller contact area, so it gives a greater pressure. Same force, different area. Evidence: Mama Wanjiru pushes with the same force of 40 N on both knives. The sharp edge (0.1 mm) has an area of $5 \times 10^{-6} \text{ m}^2$ and the blunt edge (1 mm) has $5 \times 10^{-5} \text{ m}^2$. The pressures are 8 MPa for the sharp knife and 0.8 MPa for the blunt knife, so the sharp knife gives 10 times more pressure. Reasoning: Pressure is force acting at right angles per unit area, $P = F/A$. With the force constant, a smaller area gives a larger pressure. Tomato and meat give way at about 2 MPa. The sharp knife's 8 MPa is above this, so it cuts. The blunt knife's 0.8 MPa is below it, so the food is only squashed. The difference between the knives is the area, not the force. Model revision: In Lesson 1 my model said the sharp knife has a thinner edge and touches in a smaller place. That idea was right. I added the definition $P = F/A$ and the brick results (Lesson 2), the knife calculations with the numbers and 2 MPa value (Lesson 3), and later the ideas about liquids, air and hydraulics that showed pressure, not just force, decides the effect. I also learned that the air pushes equally on both knives, so it does not explain the difference.)
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim and use of the key idea (same force, different area)	States a clear claim that the difference is the area, not the force, uses the words 'same force, different area', and links pressure to force and area correctly.	States that area is different and pressure is greater for the sharp knife, but the idea that force is the same is missing or unclear.	Claim is missing, or says the sharp knife uses more force, or gives no link to pressure.
Calculations and units	All areas and pressures are correct ($5 \times 10^{-6} \text{ m}^2$, $5 \times 10^{-5} \text{ m}^2$, 8 MPa, 0.8 MPa) with correct unit conversions and units throughout the other calculations.	Method is correct but there is one small error in conversion or units.	Several errors or missing units, or $P = F/A$ is not used correctly.
Use of evidence and threshold reasoning	Compares 8 MPa and 0.8 MPa with the 2 MPa value to explain cutting versus squashing, and supports the idea with evidence from the brick, camel and other lessons.	Uses the knife numbers but uses the 2 MPa value or other lessons' evidence only partly.	Gives little or no evidence, or does not use the 2 MPa value.
Extension to liquids, air and hydraulics	Correctly calculates and explains dam pressure, the air force on the glass and the jack, and clearly contrasts the jack (same pressure, different area) with the knives.	Calculates most of these correctly but explanations are brief or the contrast with the knives is weak.	Calculations or explanations are mostly missing or incorrect.

Model revision and communication	Compares the Lesson 1 model with the final model, naming what stayed, what changed and which lessons caused it, in clear organised sentences.	Describes some changes to the model but with few links to lessons.	Does not compare the models or the writing is unclear.
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